

Opinion and Special Articles: Remote Evaluation of Acute Vertigo

Strategies and Technological Considerations

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Abstract

Patients with acute vestibular disorders are often a diagnostic challenge for neurologists, especially when the evaluation must be conducted remotely. The clinical dilemma remains: Does the patient have a benign peripheral inner ear problem or a worrisome central vestibular disorder, such as a stroke? The use of a focused history and the virtual HINTS (head impulse test, nystagmus evaluation, and test of skew) examination are key steps towards correctly diagnosing and triaging the acute vertiginous patient. When looking for signs of vestibulo-ocular dysfunction, there are important technological and practical considerations for an effective clinical interpretation.

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Dizziness is a loosely used term that can imply different types of internal sensations. With over 4 million annual emergency department “dizzy” visits annually,¹ it is imperative to promptly differentiate between benign and sinister etiologies while avoiding unnecessary testing or diagnostic errors. The groundwork has been laid with easy-to-use video-oculography (VOG) goggles equipped with high-speed cameras and motion sensors capable of tracking eye and head positions. These portable VOG devices are key tools used by vestibular experts for “tele-dizziness” consultation services,² a true value for practices without in-house access to subspecialty care.

The restraints imposed by social distancing during the pandemic have shifted patients’ attitudes towards seeking medical care when ill. Consequently, we must rely on telehealth platforms while patients remain at home. This new norm demands a focused evaluation that can rapidly triage patients.³ This is particularly important in patients with acute vestibular symptoms, because the pressing question is whether the underlying cause is anatomically peripheral and likely benign, or central and potentially life-threatening.

Here we outline a focused practical approach that can be applied to remotely perform the clinical examination in the acutely dizzy patient, while highlighting different strategies and technological considerations.

Technological Considerations

The rapid implementation of video applications has facilitated both synchronous (online with patient and doctor interacting) and asynchronous (offline evaluation of the patient video by the doctor) encounters. The process is influenced by the device type, audiovisual platform, broadband, and the technological aptitude of the users.^{4,5} These elements determine the video quality, as they affect the resolution, frame rate, and data transfer rate of the video connections. There are strategies that patients can use to mitigate problems regarding video quality. Two simple yet crucial issues that can be addressed are the eye-to-camera distance and illumination of the eyes. In general, a well-focused close-up view of the eyes will provide higher resolutions with increased pixel density of the relevant image details.⁶ Similarly, better lighting conditions improve the quality of the images.

It is often difficult to modify the frame rate (i.e., the number of frames per second [fps]) or data transfer rate (i.e., the amount of data transmission or bit rate), as these specifications are usually fixed. Some of the best strategies to address data transfer issues include using high-speed Internet connections and avoiding platforms that only allow low data transfer rates. Geography and socioeconomic status often limit the speed of Internet connectivity and the quality of the telemedicine platforms.⁷ Preliminary simulations suggest that the frame rate of the video stream should be at least 60 fps to reliably capture abnormal eye movements such as nystagmus (figure 1).

When using smartphones/tablets, one strategy to ensure the best quality is to record from the rear-facing camera as it has a higher video frame rate. This approach may be particularly useful during asynchronous video visits or when patient assistance is readily available during live visits. For the technologically savvy user, 2 cameras can provide a close-up high-resolution/high frame rate eye movement recording (phone rear-facing camera) and a wider field of view of the patient (computer or tablet camera). This approach can be especially helpful during the evaluation of positional vertigo, as the position of the head can be seen from the camera with a wider field of view (figure 2 and video 1). Furthermore, if the pattern of positional vertigo fits with a diagnosis of benign paroxysmal positional vertigo (BPPV), one can guide the patient immediately through treatment during the remote visit. Finally, mobile applications can be used to facilitate the virtual ocular motor and vestibular examination.⁸

The Key Components of a Focused History

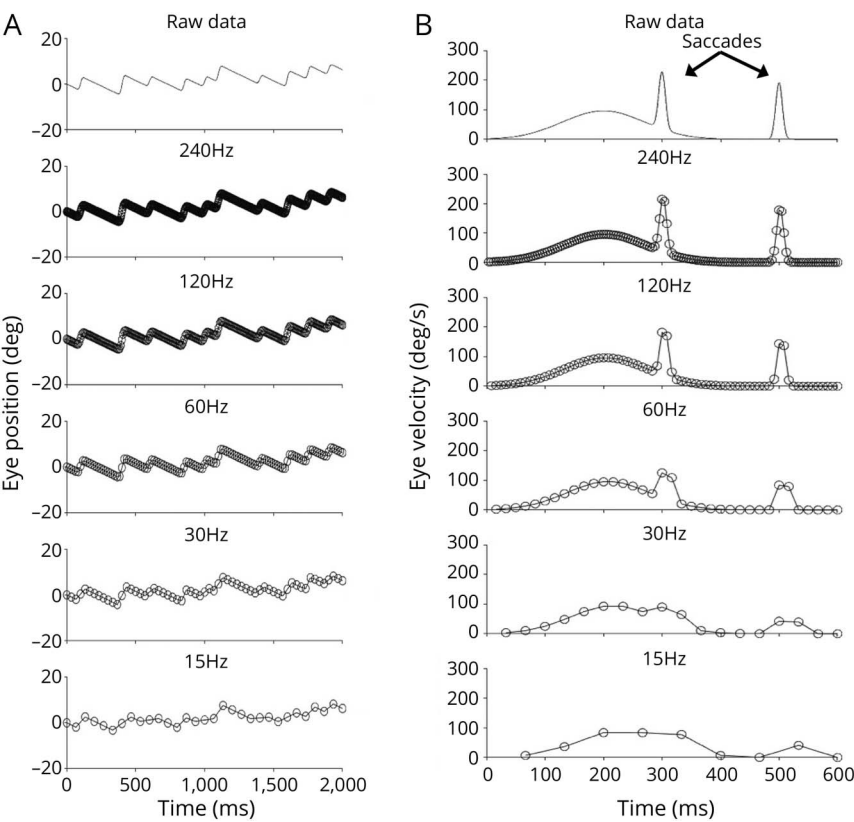
A focused history highlighting timing, triggers, and risk factors is key to the triaging process (table 1).⁹ The provider should ask whether the symptoms are intermittent or sustained and determine whether the symptoms occur spontaneously or by specific triggers; for example, if BPPV is suspected, careful questioning often confirms the affected canal.¹⁰ Age, ischemic risk factors, and the presence of associated symptoms also play an important role in triaging.⁹

The Virtual Examination

Once a focused history has been obtained and patients are appropriately triaged, a targeted virtual examination can help distinguish between peripheral and central causes (table 2).^{9,11} The general appearance of the patient may yield diagnostic information. For example, an abnormal head tilt suggesting utricular imbalance often accompanies a skew deviation, and almost always indicates a central process. A skew can be elicited using a modified alternate cover test with the patient’s palm, a close-by person, or other household item acting as an occluding apparatus (figure 2).¹² Each eye is alternately occluded every 2 to 3 seconds while the patient focuses on the center of the camera, and the examiner looks for a vertical corrective refixation movement.

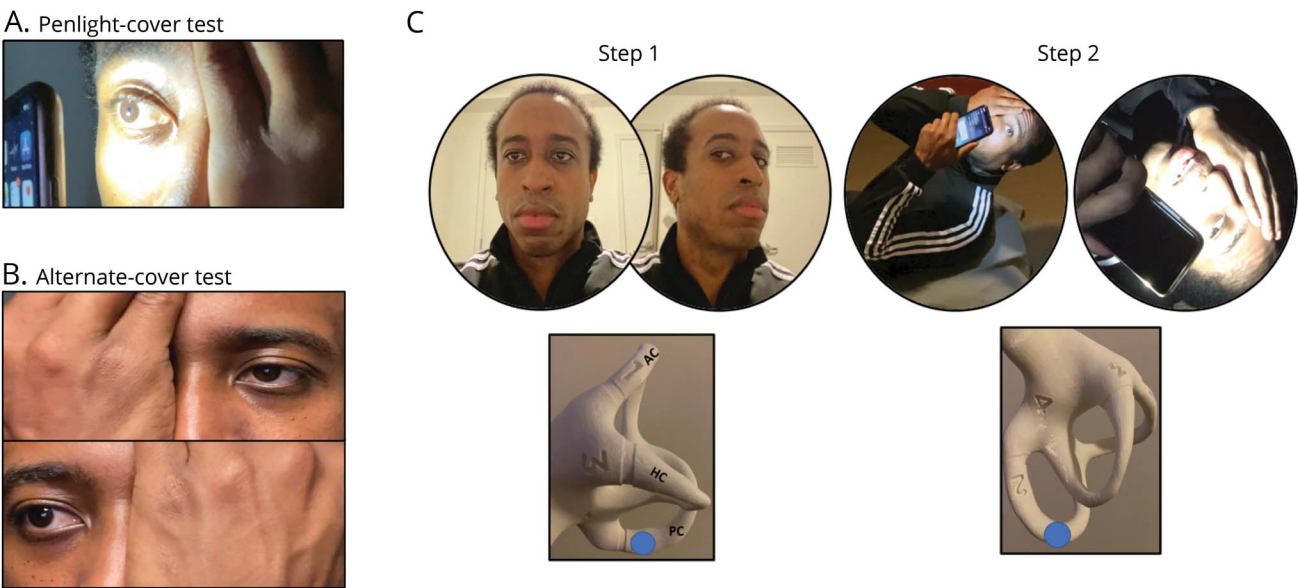
For virtual assessment of the head impulse test, the patient can be instructed to perform self-generated active head impulses on command, while fixing on a visible target on the screen (table 1 and video 2). A brief instructive demonstration by the examiner is helpful to ensure the correct size and speed of the head impulses. Again, a high enough video frame rate is needed to capture the corrective saccades (figure 1). One must also remember that “active” head impulses may be challenging to apply remotely, and can sometimes mask

Figure 1 Video Interpretation Simulation



Simulated data show the effect of frame rate on the interpretation of nystagmus (A) and corrective "catch up" saccades during the head impulse test (B). Jerk nystagmus can still be appreciated at 30 frames per second (fps). At lower frame rates, the distinction between slow phases and quick phases of the nystagmus becomes unclear. For the head impulse test, catch-up saccades appear slower and blend with the video-oculography response at the frame rates of 60 fps or less. Note that this simulated data did not include noise and real data collected from a patient will also include some level of noise that may require a higher frame rate.

Figure 2 The Virtual Examination



(A) A modified version of the penlight-cover test, where the patient's cell phone serves the dual purpose of a camera and a light source, essentially creating a makeshift video-oculography device. (B) A modified version of the alternate-cover test by performing alternating monocular occlusion using the palm for 2–3 seconds at a time at the command from provider. (C) Virtual Dix-Hallpike maneuver using the modified penlight cover test. The patient turns the head 45° to one side while seated (left side is shown here), and then moves en bloc straight back with head slightly hyperextended at about 30°. Blue dots simulate moving otoconia within the posterior semicircular canal during the maneuver. AC = anterior canal; HC = horizontal canal; PC = posterior canal.

Table 1 Key History and Examination Findings in the Main 5 Differential Diagnoses of an Acute Vertigo Presentation

Diagnosis	Key findings	
	History	Virtual examination
Vestibular neuritis	• AVS • No hearing symptoms • Preceding viral illness	• Spontaneous unilateral mixed horizontal-torsional nystagmus that increases with fixation removal • Contralaterally directed corrective saccades with active HIT
Stroke	• AVS • Associated neurologic symptoms • Acute unilateral hearing loss/tinnitus • Vascular risk factors	• Direction-changing, gaze-evoked, or pure vertical/torsional nystagmus • Skew deviation or head tilt • Normal or abnormal HIT • Unilateral hearing loss
Vestibular migraines	• Episodic vertigo (spontaneous or triggered) • Headache plus migrainous features • Aural symptoms	• Normal examination or persistent positional nystagmus
Ménière disease	• Recurrent spontaneous episodic vertigo • Fluctuating hearing loss ± other aural symptoms • Lermoyez syndrome (hearing improves as vertigo begins) • Drop attacks	• Spontaneous mixed horizontal-torsional nystagmus that may change direction
BPPV	• Transient positional vertigo	• Upbeat-torsional nystagmus (pcBPPV); geotropic or apogeotropic nystagmus (hcBPPV)

Abbreviations: AVS = acute vestibular syndrome; BPPV = benign paroxysmal positional vertigo; hcBPPV = horizontal canal benign paroxysmal positional vertigo; HIT = head impulse test; pcBPPV = posterior canal benign paroxysmal positional vertigo.

vestibular hypofunction.^{13,14} If available, mobile-based quantitative VOG may provide a more accurate assessment.⁸

Spontaneous jerk nystagmus is the hallmark of vestibular imbalance and is best appreciated when the patient is close to the device with eyes fully illuminated and focusing on the camera. Because peripheral nystagmus becomes more intense with gaze in the direction of quick phase (Alexander law), and characteristically increases or only appears when visual fixation is eliminated,¹⁵ a simple strategy to remove fixation is to ask the patient to close the eyes and observe for movement of the corneal bulges.¹⁵ The effect of fixation on nystagmus can also be simulated using a modified version of the penlight cover test.¹⁶ In this modified technique, the phone's flashlight

serves the dual role of camera and light source, while the patient or a family member occludes the other eye (figure 2 and video 3). When BPPV is suspected, the patient can be instructed to perform positional maneuvers while a smartphone/tablet is held close to the eyes, either by the patient or a family member (figure 2 and video 1). When possible, the modified penlight cover test can be applied to better appreciate the nystagmus. The velocity of the nystagmus also affects one's ability to appreciate it remotely, therefore video frame rate and Internet connectivity become extremely important. The severity of symptoms and the availability of assistance during the evaluation can also affect the usefulness of these techniques. During asynchronous video visits, the lack of immediate guidance from the examiner may limit its usefulness.

Table 2 Step-by-Step Instruction for the Virtual HINTS (Head Impulse Test, Nystagmus Evaluation, and Test of Skew) Examination

Examination	Description
Command-driven active head impulse test	• Instruct patient to move close to the camera with eyes fully illuminated • Provider directs patient to fixate on a visual target on the device and perform head impulse test in the yaw plane at the direction of the provider
Nystagmus	Fixation • Instruct patient to move close to the camera with eyes fully illuminated • Instruct patient to stare straight ahead at the camera, then assess for nystagmus (for down and down/lateral gazes, instruct patient/family member to elevate eyelids to ensure visualization of the eyes) Fixation removal Modified penlight cover test: • Instruct patient/family member to perform monocular occlusion using palm • Instruct patient/family member to remove visual fixation by shining a flashlight (preferably a cellphone flashlight) in the eye opposite or • Evaluate nystagmus under closed eyelids by looking at the corneal bulges
Alternate-cover test (test of skew)	• Instruct patient to move close and focus on the camera (fixation target) with eyes fully illuminated • Instruct patient/family member to perform alternating monocular occlusion using palm for 2–3 seconds at a time at the command from provider

Hearing evaluation and assessment of balance combined with the 3-step HINTS (Head Impulse test, Nystagmus evaluation, and Test of Skew) or HINTS plus have a higher sensitivity and specificity than neuroimaging for detecting strokes.^{17,18} A crude Weber test can be performed by the patient tapping the incisors then reporting whether the sound is equal or louder from one ear. There are also various app-based screening tools (e.g., the hearWHO app [WHO, Geneva, Switzerland]) that can be deployed. Self-assessment of hearing is often unreliable.

Both the virtual HINTS and Dix-Hallpike examinations can be readily applied via telemedicine. For the most effective evaluation, the video quality, lighting condition, and distance from the recording device are the most important considerations. Further studies validating the clinical utility of these techniques will be necessary for standardization and to spur the development of virtual and augmented reality solutions for more realistic telemedicine assessments.

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